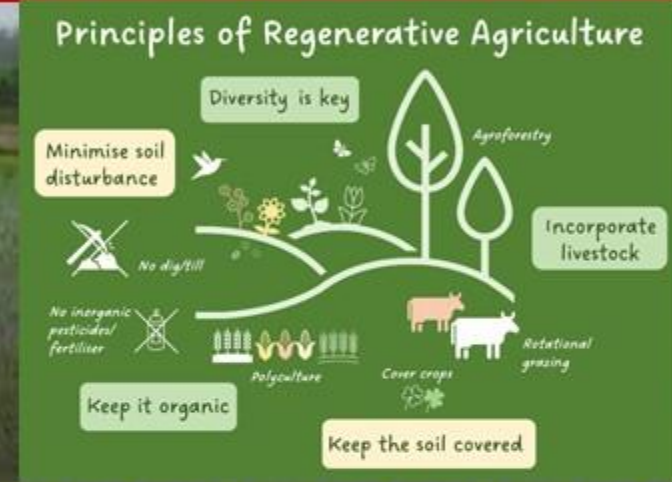


INDIAN ECONOMY

LEVEL
II



LECTURES

- FATIGUE DUE TO GREEN REVOLUTION
- REDUCTION OF SOIL ORGANIC CARBON
- CONCEPT OF REGENERATIVE AGRICULTURE

MODULE - 70



WHAT WAS THE METHODOLOGY FOLLOWED IN GREEN REVOLUTION?

6

Workers are replaced by machines, wherever possible to increase the outputs more efficiently.

5

Monocropping on fields was necessary to apply all appropriate inputs - seeds, fertilizer, water, etc., on scale and it increased the efficiency in application of inputs.



1

It applied scientific breakthrough with methods of management to obtain economies through scale.

2

To produce chemical fertilizers on large scale at low cost, large fertilizer factories were set-up.

3

Large dams and irrigation systems were also developed to feed water on a large scale.

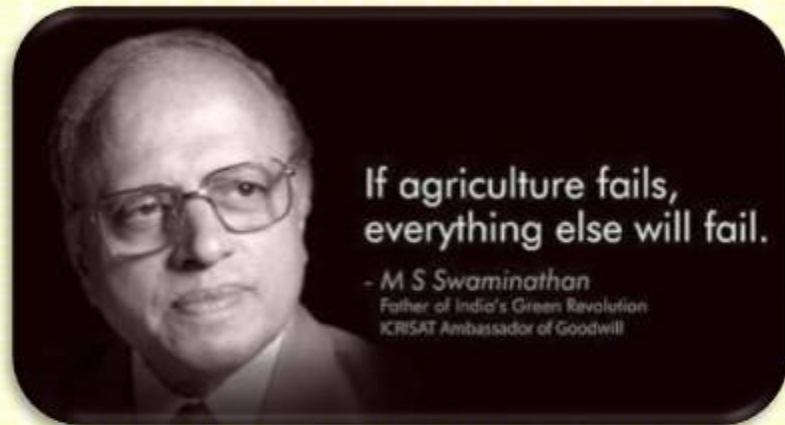
4

It focused only few crops at a time and in turn avoided diversion of land use to other non-essential crops.

FOOD SECURITY
WAS THE
FOREMOST
OBJECTIVE!



FATIGUE OF “GREEN REVOLUTION” IS CLEARLY EVIDENT!



- ‘Green Revolution’ was the term coined for the process of productivity enhancement.
- The Green Revolution in the 1960s and 1970s was based on the development and spread of new genetic strains of wheat, rice, maize and other crops characterised by their ability to utilise water, sunlight and plant nutrients effectively and convert them into grains. This was confined to areas with assured irrigation, and north-west India is mostly benefitted.
- However, even here, farming is becoming unremunerative due to adverse ecological and economic factors, with farmer indebtedness on the rise. The challenge now is to fight and overcome the fatigue of the Green Revolution.

As per one estimate, farming is no longer remunerative and over 40% of farmers would like to quit, if they have an option!

CHEMICAL FARMING RESULTS IN DEPLETION OF SOIL ORGANIC CARBON!



- Chemical farming, and even traditional manure, utilise the carbon naturally available in soil.
- This leads to initial increase in output but, subsequently carbon level gets depleted and demands more and more resources as the years pass by.
- A few years ago, ICAR Director General stated that the depletion of organic nutrients in the soil had reached alarming proportions. The acceptable level of soil organic carbon is 3-5%, but, in soils of Punjab, it has gone below 1%.
- The indiscriminate use of inorganic fertilizers in wrong combinations rendered the soil incapable of regenerating itself.
- Falling moisture retention abilities also leads to indiscriminate use of ground water.



WHAT IS SOIL ORGANIC CARBON?

- 1** It is a measurable component of soil organic matter.
- 2** Soil organic matter is composed of Carbon, Hydrogen and Oxygen. It also has small amounts of nutrients such as Nitrogen, Phosphorus, Sulphur, Potassium etc.
- 3** Organic matter makes up just around 2 to 10% of the soil mass, but has a critical role in the physical, chemical and biological functions of agricultural soils.

IN A NUT SHELL, WHAT ARE THE REASONS FOR THE REDUCTION OF SOIL ORGANIC CARBON?



Improper agriculture practices are cited as the major reasons for reduction in the soil organic carbon.

- 1 There may be natural factors such as floods, volcanoes and earthquakes.
- 2 ➤ The human induced factors are ...
 - ✓ Over grazing by cattle
 - ✓ Urban expansion
 - ✓ Improper management of industrial waste
 - ✓ Land degradation due to inappropriate agriculture practices
 - ✓ Excessive tillage
 - ✓ Frequent cropping
 - ✓ Poor irrigation and water management
 - ✓ Unscientific rotation of crops

WHAT IS THE ROLE OF SOIL ORGANIC CARBON?



- 1 It plays a key role in maintaining soil fertility by holding Nitrogen, Phosphorus and other nutrients for plant growth.
- 2 It holds soil particles together as stable aggregates, improving the soil properties such as water holding capacity.
- 3 It provides gaseous exchange under root growth.
- 4 It plays an important role as food source for soil fauna and flora.
- 5 It is acting as a buffer against toxic and harmful substances with absorption of toxic and heavy metals.

INCREASE IN SOIL ORGANIC CARBON IS IMPERATIVE!

1 FROM FARMERS' PERSPECTIVE

- Healthier soils normally have higher organic matter.
- For every 1% increase of carbon in the soil, the soil holds 20,000 gallons of water per acre. It means a greatly reduced risk of droughts. Farmers can become productive with better yields.



2 FROM CLIMATE CHANGE PERSPECTIVE

- If we are able to increase 0.4% of soil organic carbon in the top 1 metre of global agricultural soils, the carbon sequestered would offset up to 35% of greenhouse gas emissions.
- In terms of climate change mitigation, this would buy time around 10-20 years.



“4 PER 1000” IS THE INITIATIVE OF UNFCCC



- The "4 per 1000" initiative intends to increase soil organic matter and carbon sequestration through the implementation of agricultural practices adapted to local environmental, social and economic conditions, such as agro-ecology, agro-forestry, conservation agriculture or landscape management.
- This initiative is part of the Global Climate Action Agenda (GCAA) adopted by the UNFCCC at COP 22.
- This requires a shift from industrial agriculture to regenerative farming.

KEEPING HARMONY WITH THE NATURE ... THE ESSENCE OF REGENERATIVE AGRICULTURE!



- Soil has billions of lives.
- Micro-organisms and a huge amount of ecosystem services work for us.
- If we take care of the soil, the soil will take care of us.
- We should treat the soil, the plants and thriving lives in it with respect - nurturing these and ensuring that they can grow into the next generation.
- If we appreciate all life, it grows in abundance and rejuvenate us.
- Regenerative agriculture with respect to harmony is the need of the hour.

ALLOWING NATURAL ECOSYSTEMS TO FLOURISH IS THE CONCEPT OF REGENERATIVE AGRICULTURE!



Allowing natural ecosystems to flourish with whole systems thinking is the concept of regenerative agriculture.

- Regenerative agriculture is based on plants absorbing atmospheric carbon and depositing this in the soil.
- Overall regenerative practices keep plants healthy, which supports microbial activity, decomposing organic matter and storing more soil carbon.
- The core point is to try and produce more life than we take.
- It tries to restore and increase life in terms of biodiversity of plants, animals, and birds by seeking harmony with nature.

EVERGREEN AGRICULTURE BEING PROMOTED BY Dr. M. S. SWAMINATHAN ... IS NOTHING BUT REGENERATIVE AGRICULTURE!

- While organic farming excludes the use of mineral fertilisers, chemical pesticides and genetically modified crops, green agriculture, practiced widely in China, is based on the principle of integrated pest and nutrient management, alongside integrated natural resources conservation and enhancement.
- It is promoting a diversity of sustainable agricultural practices, suited to different agro-ecological and social environments, in both rainfed and irrigated areas.
- It is also adopting a “de-ecology” approach to ensure that ecologically sustainable farming technologies become available.

- A de-ecology approach involving, for instance, mandatory rainwater harvesting and the adoption of a three-year rotation, in which a legume is introduced, both during kharif and rabi seasons that will help promote a win-win situation both for farm families and the country.



FOR REALISING THE GOAL OF EVERGREEN AGRICULTURE ...

SUSTAINABLE AGRICULTURE PRACTICES HOLDS THE KEY!

- Examples of Sustainable Agriculture Practices (SAPs)
 - ✓ Organic Farming (Sikkim).
 - ✓ Natural Farming (ZBNF).
 - ✓ Integrated Farming Systems.
 - ✓ Agroforestry.
 - ✓ Precision Farming.

These could be economically remunerative, socially inclusive and environmentally benign!



... BUT, THESE ARE YET TO BE EMBRACED BY THE INDIAN FARMER!

- Sustainable agriculture remains at the margins. No single SAP has been adopted by more than 4% of farmers.
- Support from the Centre is limited.
- India's National Mission on Sustainable Agriculture receives only 0.8% of the agricultural budget of Rs.1.42 lakh Cr in 2022-23.

The lack of state support limits the mainstreaming of sustainable agriculture.



SUGGESTIONS TO SCALE-UP SUSTAINABLE AGRICULTURE PRACTICES

1 KNOWLEDGE EXCHANGE AND CAPACITY BUILDING

- There should be support for knowledge exchange and capacity building.
- Many of the practices are knowledge- and skill-intensive. Farmers need handholding in the early phases.
- To accelerate the process, Government must leverage the presence of more than 1,000 civil society organizations that are already promoting farmer-to-farmer capacity building for sustainable agriculture.

2 TECHNOLOGY INNOVATION AND ADOPTION

- There is a need to support technology innovation and adoption to mechanize labour intensive activities associated with SAPs.
- We must incentivize innovators and entrepreneurs through channels like the AIM to encourage the development of farm implements for SAPs.
- Alongside, we should also support local micro-businesses through state livelihood missions to produce and sell ready-made inputs such as vermicompost and organic fertilizers.

SUGGESTIONS TO SCALE-UP SUSTAINABLE AGRICULTURE PRACTICES

3 INCENTIVIZE OUTCOMES

- At present, the government is giving input-based subsidies for fertilizer and power. It is better, if the government incentivize outcomes such as annual nutrition output per hectare and enhanced ecosystem services such as water conservation, or desertification reversed.
- Outcome-based support could encourage innovation among farmers and allow the adoption of alternative approaches, including SAPs.

4 R & D SUPPORT AND PROMOTING IMPACT STUDIES

- We must significantly enhance research and development support for SAPs.
- We should also promote impact studies, comparing these with conventional farming practices across agro-climatic zones.
- Focusing on landscape-level long-term impact studies could further scale-up SAPs even in irrigated areas.
- The regular comparisons of outcomes, amid climate change, is important for safeguarding farm incomes, nutrition security and natural capital.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

98499 42299